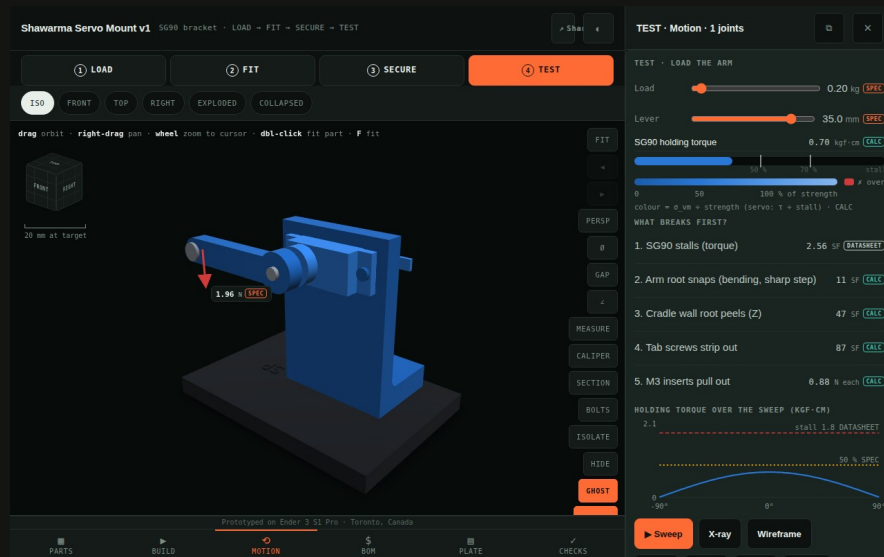


AV v8

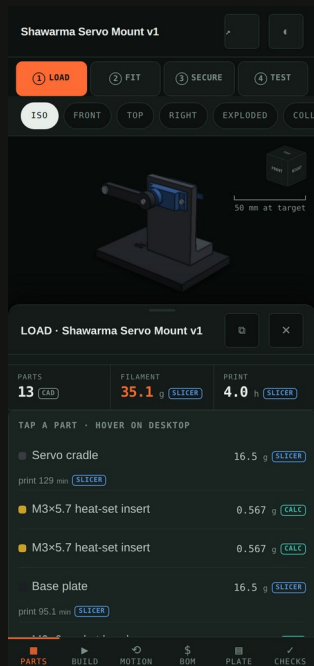
See if it fits, bolts up and holds — before you print it

Shawarma Servo Mount v1 · LOAD → FIT → SECURE → TEST · one link, every number with its source



TEST phase: 2.0 kg on the arm — the SG90 stalls first (red), the arm root carries the stress.

Four taps tell the whole story



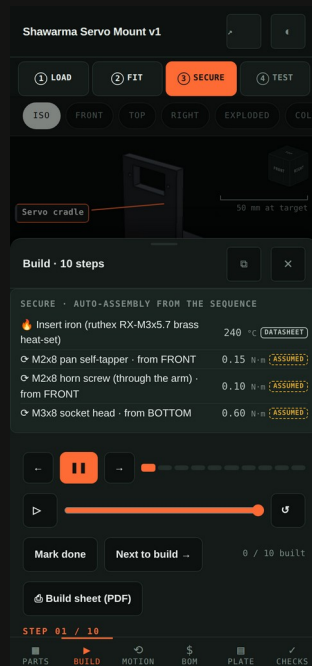
1 LOAD

mass + print time, from G-code



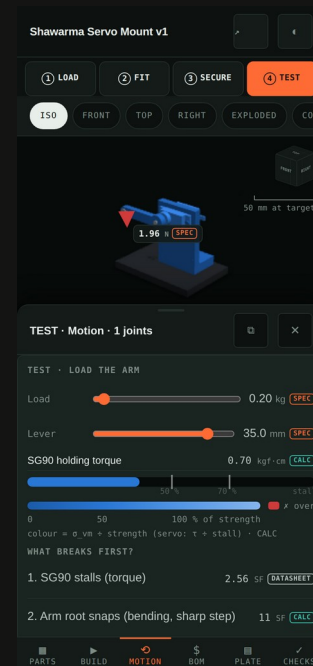
2 FIT

holes coloured by how YOUR printer prints



3 SECURE

auto-build: 240 °C inserts, screws from their side



4 TEST

drag the load, see what breaks first

The servo is the fuse

0.51 kg

SG90 stalls

1.8 kgf·cm DATASHEET ÷ 35
mm lever

2.24 kg

**arm root snaps,
sharp step**

Kt 2.43 (Peterson) · CALC

4.05 kg

with a 2 mm fillet

Kt 1.34 · CALC

0.10 %

**PyBullet vs hand
calc**

inverse-dynamics torque

The cheap part gives up first, safely — the bracket outlives the servo by 4.4× (arm SF at stall).

Caught before a gram was printed

Tips over

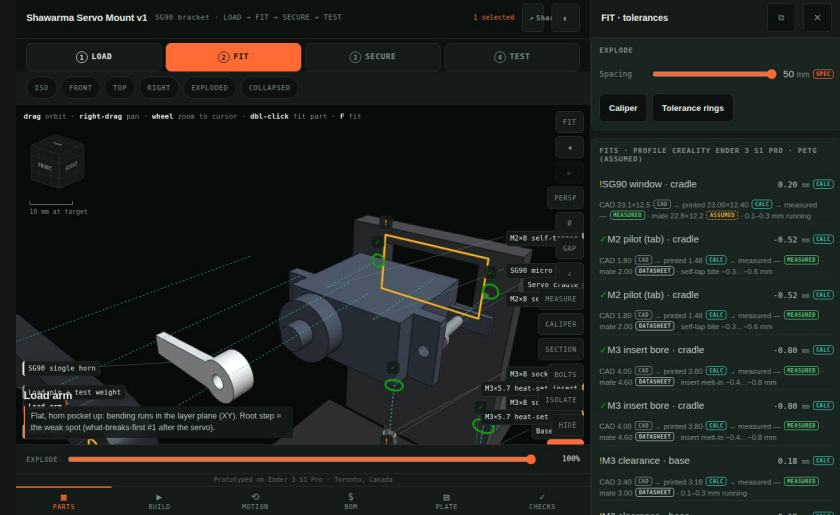
CoG 8.0 mm outside the base with 200 g → base under the load: 12.1 mm inside

Pilot too tight

Ø1.5 prints Ø1.13 (split risk) → Ø1.8 CAD prints Ø1.48: a clean M2 bite

SG90 may not fit

datasheets disagree by 0.8 mm → worst-case window -0.10 mm: calipers first



FIT: rings coloured from the printer's tolerance profile (green snug · yellow running · red sloppy/tight).

Measure three things, then sell it

- Caliper the SG90: body, tab span, hole pitch
- Print + measure the hole coupon (9 holes, 46 min)
- Weigh the parts; read the spool's datasheet

12 UNVERIFIED tiles turn PASS or FAIL — both are useful.

AV for your build

\$450 — LOAD + FIT + SECURE

\$1,200 — + TEST, checks,
MEASURE_ME

First one free for a testimonial